Reliability Engineering and System Safety 202 (2020) 107067



Contents lists available at ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress



Joint optimization of safety barriers for enhancing business continuity of nuclear power plants against steam generator tube ruptures accidents



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ARTICLE INFO

Keywords:
Business continuity management (BCM)
Safety barrier
Joint optimization
Event tree (ET)
Mixed-integer genetic algorithm (MIGA)
Nuclear power plant (NPP)
Steam generator tube rupture (SGTR)

ABSTRACT

In nuclear power plants (NPPs), different types of safety barriers are designed to ensure the safe and continuous operation of the NPP against disruptive events. These safety barriers, although designed to operate in different phases of the accidents evolution, are often optimized separately, without considering their collective effects on preventing disruptions and quickly recovering from the disruptions. This paper develops a joint optimization model for synthetically optimizing safety barriers of different natures, including prevention, mitigation, emergency and recovery barriers to enhance the business continuity of the NPP, considering the threat of steam generator tube rupture (SGTR) accidents. The joint optimization is guided by a business continuity metric called expected business continuity value (EBCV). A physics-of-failure model is developed to describe the crack growth process of the steam generator tube and to model the effect of the prevention barriers, i.e., periodical inspection of the crack length. An event tree model is developed to describe the evolution of the SGTR-initiated accident and to model the effect of the mitigation and emergency barriers. Recovery measures are also considered via a widely-used logarithmic function model. A mixed-integer genetic algorithm (MIGA) is used to obtain optimal solutions of the joint optimization model. The results show that the developed joint optimization model can achieve better performance in terms of business continuity, compared to the conventional methods that optimize the safety barriers separately.

1. Introduction

Steam generator (SG) is a heat-exchanging system that transfers heat from the primary loop to the secondary loop in a pressurized water reactor (PWR) to produce steam to drive the turbines [1]. One relevant safety issue in PWR is the rupture of SG tubes, known as steam generator tube rupture (SGTR), which can be an accident initialing event induced by a crack growth process in the SG tube [2]. SGTR accidents can cause severe consequences related to the leakage of radioactive materials [3]. The safe operation of nuclear power plants (NPPs) is ensured by a suite of safety barriers designed to prevent undesired events or accidents, and contain or mitigate their consequences when they occur [4, 5].

Prevention barriers are designed to work in the pre-accident or pre-failure phases and aim at reducing the probability of occurrence of accidents [6, 7]. In the case of SGTR accidents, one commonly adopted prevention barriers is to make periodical inspection and timely preventive plugging of the defective tubes [8]. Mitigation and emergency barriers intervene after the accident initiating event occurs and aim at containing the evolution of the accident so that its consequences can be minimized. Examples of mitigation and emergency barriers for SGTR accidents include the reactor trip system (RTS), reactor depressurization system (RDS), refueling water storage tank (RWST), reactor cooling system (RCS), etc. [9]. Recovery barriers aim at restoring to the normal operation system functionality timely after the accident [10]. For example, in the event of a SGTR, the recovery measures could include replacing the ruptured tube, cleaning up the contaminated area (if any), etc. [11, 12].

In practice, the different safety barriers are designed separately for optimal performance, considering the constraints of limited resources [13]. For instance, [14] has proposed the optimal design of risk-based inspections in power and process plants. In [15], an enhanced preventive maintenance optimization model based on a three-stage failure process has been proposed for NPP components. In [16], condition-

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https://doi.org/10.1016/j.ress.2020.107067
Received 25 November 2019; Received in revised form 29 May 2020; Accepted 3 June 2020
Available online 10 June 2020
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Acronyms

BCM	business continuity management
EBCV	expected business continuity value
ET	event tree
MIGA	mixed-integer genetic algorithm
NPP	nuclear power plant
PDF	probability density function
PSA	probabilistic safety assessment
RDS	reactor depressurization system
RTS	reactor trip system
RWST	refueling water storage tank
SG	steam generator
SGTR	steam generator tube rupture

Notation

a	crack length
c_e	cost-effective parameter for $II-$th consequence
C	production capacity of NPP
C_i	consequence with $i-$th severity
C_P	cost in preventive phase
C_M	cost in mitigation phase
C_R	cost in recovery phase
C_{plug}	unit price for plugging one tube
C_{th}	total budget
$\frac{da}{dt}$	crack growth rate
ΔK	stress intensity factor
$L([0, T])$	potential loss in $[0, T]$
$L_d([0, T])$	direct loss in $[0, T]$
$L_{in}([0, T])$	indirect loss in $[0, T]$
L_{tol}	maximum tolerable loss
n_{tube}	number of tubes
PF	vector representing the baseline value for the failure probabilities of the mitigation measures
$p_{rup,\ tube}$	probability of one tube rupture
T	estimation horizon for business continuity assessment
T_{recv,C_i}	recovery time for the $i-$th consequence
μ	relationship between cost-effectiveness parameters of C_{II} and C_{III}

based maintenance optimization for deteriorating systems has been investigated by considering inspection intervals and preventive maintenance thresholds as decision variables. Only prevention activities were taken into account in these works. There are also a number of works considering the optimal design of emergency and mitigation barriers. For example, a simplified probabilistic safety assessment (PSA) and a reliability allocation model have been developed to improve the safety level of PWR by the optimal allocation of redundancies of the emergency and mitigation barriers [17]. In [18, 19], a redundancy allocation model for series-parallel systems has been used to improve the reliability of mitigation measures. In [20], an optimization problem is formulated and solved for the minimum average total cost of nuclear fail-safe systems, where the average total system cost was subject to a restricted type I design error. The optimization of recovery measures has also been considered in the literature. For example, in [21], economically optimal strategies for recovering from a NPP accident have been discussed. In [22], joint restoration processes for multiple systems have been modeled and the effectiveness of five different restoration strategies with respect to resilience has been compared. A resilience-based optimization methodology has been proposed in [23].

Most existing research works, as reviewed above, optimize the safety barriers separately. In practical problems, however, the safety barriers at different phases need to work jointly to ensure that the NPP can be operated continuously and safely. In this paper, we propose a joint optimization model to ensure that the different safety barriers can achieve holistic optimal performances. The performance of the safety barriers system is quantified through the concept of business continuity. Defined as "the holistic management process that identifies potential threats to an organization and the impacts to business operations those threats, if realized, might cause, and which provides a framework for building organizational resilience with the capability of an effective response that safeguards the interest of its key stakeholders, reputation, brand and value-creating activities" [24], business continuity management (BCM) has received more and more attention in recent years, as a holistic requirement on the overall performance of a system [25, 26]. A loss-based business continuity metric was defined in [5] for business continuity assessment of an oil tank farm.

It is worth to note that both business continuity and resilience are related to the capability of a system to handle disruptions and require the system to prepare for disruptions, contain damages and quickly recover from disruptions. In this sense, they are very similar concepts. However, some major differences do exist between them, both conceptually and mathematically. Conceptually speaking, the focus of business continuity is on the entire disruptive handling process, including both pre-disruption prevention, mitigation and emergency response during disruption and post-disruption recovery. Resilience, on the other hand, focuses primarily on system responses after the disruptive events occur. Mathematically speaking, the business continuity is measured based on a numerical metric called business continuity value in this paper. The business continuity value gives an estimation of the potential financial losses caused by the disruptive events and can be used to quantify the financial risk of business failures due to disruptions. Existing resilience metrics, e.g., the resilience triangle, are mainly based on system performance, but do not directly relate to financial losses directly.

In this work, we adopt the quantitative business continuity metric in [5] to guide the joint optimization of the pre- and post-disruption barriers, and use a mixed-integer genetic algorithm (MIGA) [27] to derive the optimal solution. The contributions of this work can be summarized as follows:

(1) A joint optimization model is proposed to enhance the business continuity of NPP.
(2) MIGA is implemented to solve the joint optimization problem.

The remaining of the paper is structured as follows. Section 2 formally defines the problem. Section 3 illustrates the SGTR event and the corresponding safety barriers at different stages. Section 4 provides a joint optimization model and a solution method for enhancing business continuity of the NPP. Section 5 performs a sensitivity analysis of the parameters in the four stages. Finally, Section 6 concludes this work.

2. Problem description

One SG of a PWR is considered, which is equipped with a bundle of 3592 inverted U tubes. Each U tube has an outside diameter Gaussian distributed with a mean of 22.23 mm and a standard deviation 0.1667 mm; and a thickness Gaussian distributed with a mean of 1.27 mm and a standard deviation of 0.0592 mm. The tubes are subject to different degradation mechanisms like stress corrosion cracking (SCC), fatigue, pitting corrosion and fretting wear. A list of the NPP parameter values is presented in Table 1 [8].

Different safety barriers are presented for preventing SGTR, containing its consequences and recovering from the possible disruptions

Table 1.
Parameters of the NPP.

Parameter	Value
Generation capacity of NPP (C)	1100 Mwh
Number of tubes (n_{tube})	3592
Tube outer diameter (d)	N (22.23,0.1667) mm
Tube thickness (b)	N (1.27, 0.0592) mm

caused. Table 2 summarizes the safety measures considered in this paper and highlights the category they belong to: prevention, mitigation, emergency and recovery. In this paper, the improvement measures for the safety barriers in mitigation and emergency phases are the same, i.e., adding redundancies. Thus, to simplify the analysis, we do not further distinguish between the mitigation and emergency phases.

3. Modeling the individual safety barriers

In this section, we present the model of the performance of each safety barrier and the associated costs. Prevention safety barriers are discussed in Section 3.1, followed by mitigation and emergency barriers in Section 3.2, and recovery measures in Section 3.3.

3.1. Modeling prevention safety barriers

SGTR is the break of one or more SG tubes, which can be caused by different degradation mechanisms, e.g., stress corrosion cracking (SCC), fatigue, pitting corrosion, fretting wear [28]. As reported in [3], a fraction of 60% ~ 80% of SGTR events is caused by SCC. For this reason, without loss of generality, in this work, the SGTR is considered to be due only to SCC. We did not consider multiple degradation processes and coupling failure processes in this paper. However, the proposed joint optimization model can be naturally extended to account for these phenomena. For instance, if we consider SCC and pitting corrosion together in the joint optimization model, the optimization model in Eq. (17) is still valid. We only need to update the probability of SGTR in Eq. (8), considering all the possible failure mechanisms and dependencies among them. The main prevention safety measure is to inspect the tube periodically and plug the dangerous tubes when necessary.

The tube SCC growth process can be divided into two stages: onset and propagation of cracks inside the tube wall [29], as shown in Fig. 1. During the onset phase, the crack grows slowly; then, when the crack size reaches a critical point, it begins to grow more rapidly in the propagation phase. The SCC process is often modelled by a two-stage physics-based crack growth model. In the two-stage model, it is assumed that the critical crack length beyond which the process enters the propagation phase (see Fig. 1) is 0.1 mm. The duration of the onset phase is assumed to follow a lognormal distribution with parameters μ_c and σ_c, where μ_c denotes the mean value and σ_c is the standard deviation [8]. Let $t_{critical}$ represent the time needed for the crack to reach the length of critical size 0.1 mm. Then, we have:

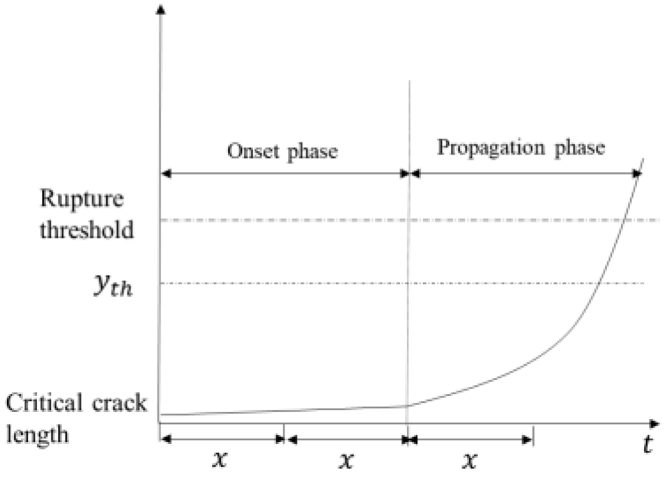


Fig. 1. Tube crack growth process.

$$t_{critical} \sim \log normal(\mu_c, \sigma_c^2), \tag{1}$$

The critical crack occurs at about 9.3 years (with standard deviation of 3.162 years) at the operating temperature of 330 °C [8]. Thus, in this paper, we take the values of the parameters as $\mu_c = 9.3$ years, $\sigma_c = 3.162$ years.

The Scott model is often used to model the propagation phase of the crack growth process (see Fig. 1), in which the crack growth rate is empirically modelled as a function of stress [30]:

$$\frac{da}{dt} = \alpha(K - K_{th})^m, \tag{2}$$

$$K = F\sigma\sqrt{\frac{\pi a}{2}}, \tag{3}$$

$$\sigma = \frac{\Delta P \cdot d}{2b} \tag{4}$$

where $\frac{da}{dt}$ is the crack growth rate, a is the crack length, α, K_{th} and m are constant parameters related to the component material properties, σ is the stress at the crack tip, ΔP denotes the pressure difference. In this paper, the material of steam generator tubes is assumed to be Alloy 600. Based on the material properties, the values for the parameters in Eqs. (2), (3) can be determined, as shown in Table 3.

Based on the physical model of the crack growth process, the effectiveness of the prevention safety measures can be further modeled. It is assumed that periodical inspections are conducted every x months and during each inspection, the crack size can be measured through techniques like eddy current testing [31], ultrasonic testing [32], etc. If the measured crack length is beyond a given preventive maintenance threshold, denoted by y_{th}, the corresponding tube is plugged to prevent further damages. The probability of SGTR, denoted by $p_{rupt,\ tube}$, is used to represent the performance of the prevention safety barrier: the higher the value of $p_{rupt,\ tube}$, the worse the performance of the

Table 2.
Safety barriers considered in this study.

Safety barrier	Category	Function description
Periodic inspection and maintenance	Prevention	Periodically inspect the tubes and timely plug those defective tubes whose crack length is beyond the maintenance threshold.
Reactor trip system (RTS)	Mitigation	When the reactor power exceeds a given safety operating limit, the RTS automatically shut down the reactor, in order to prevent core damage.
Reactor depressurization system (RDS)	Mitigation	When a loss-of-coolant event is caused by the tube rupture, the RTS will work to prevent over-pressurization of the reactor vessel.
Refueling water storage tank (RWST)	Mitigation	Store cooling water for emergency cooling of the reactor core.
Reactor cooling system (RCS)	Mitigation	If the reactor fails to scram, RCS will pump water into the reactor for emergency cooling.
Repair of the damages caused by the SGTR	Recovery	Replace the ruptured tubes and restore the plant to normal operation.

Table 3.
Parameter values of crack growth model.

Parameter	Value
a_0	0.1
F	0.93
m	1.16
K_{th}	9
ΔP	N (8.3,0.33) Mpa

prevention barrier. In our model, the probability of SGTR is formulated as a function of x and y_{th}:

$$p_{rupt,tube} = f(x, y_{th}). \tag{5}$$

Monte Carlo simulation is used to evaluate the value of $p_{rupt.tube}$, as shown in Algorithm 1, where we can see the simulation starts from setting parameter initial values. Afterwards, we simulate the crack growth process and get the critical time point when the crack length reaching the critical point 0.1 mm. This process is repeated times. Then, we further simulate crack propagation phase to find the rupture time. If an inspection was done before the rupture time, rupture is prevented. Otherwise, a rupture occurs. The rupture probability can, then, be easily calculated based on the MC samples. 10^7 Monte Carlo simulation are carried out to simulate the crack growth process in this work, namely $N_s = 10^7$ in Algorithm 1. where T is the calculation time horizon; a_{th} represents the rupture threshold; t_{rup} is the rupture time; a_{in} implies the crack size measured at inspection.

The cost spent in the prevention phase, represented by C_P, is a function of inspection interval x and plugging rate p_{plug}:

$$\begin{aligned} C_P &= C_{insp} \cdot n_{insp} + n_{tube} \cdot p_{plug} \cdot C_{plug} \\ &= \frac{T}{x} \cdot C_{insp} + n_{tube} \cdot p_{plug} \cdot C_{plug}, \end{aligned} \tag{6}$$

where C_{insp} denotes the cost of a single inspection of the tube; C_{plug} is the unit price for plugging one tube; and p_{plug} is the plugging rate of the tube, i.e., the fraction of tubes being plugged. The value of p_{plug} depends on the values of x and y_{th}, as shown in Algorithm 1.

3.2. Modeling mitigation and emergency barriers

As reported in Table 2, the emergency and mitigation barriers include the RTS, RDS, RWST and RCS. Their effects on containing the consequence of SGTR can be modelled using an event tree (ET), as shown in Fig. 2.

Depending on the performance of the different barriers, 6 categories of consequences i.e., C_1, C_2, ⋯, C_6, can occur following the SGTR. These consequences can be grouped into three macro-categories based on their severity, as in Table 4. Based on the ET in Fig. 2, the occurrence probabilities for $C_1 \sim C_6$ can be quantified as a function of the event probabilities. Conceptually, we denote these by:

$$p_{C_i} = f_{ET}(p_{SGTR}, p_1, p_2, p_3, p_4) \tag{7}$$

where p_{SGTR} is the occurrence probability of SGTR and p_1, p_2, p_3, p_4 represent the failure probability of component in RTS, RDS, RWST and RCS, respectively. It should be noted that SGTR occurs when at least one tube ruptures in the SG. Thus, p_{SGTR} can be calculated from:

$$p_{SGTR} = 1 - (1 - p_{rupt,tube})^{n_{tube}} \tag{8}$$

Algorithm 1
. $p_{rup.tube}$ calculation based on Monte Carlo simulation.

Algorithm 1: $p_{rup.tube}$ calculation based on Monte Carlo simulation.
Inputs: $x, d, \Delta P, b, a_0, F, m, K_{th}, a_{rup.th}, T, N_s$;
Outputs: $p_{rupt,tube}$;
$t_{in} = [x, 2x, 3x, \cdots, \boldsymbol{kx}], \boldsymbol{k} = 1, 2, \cdots \boldsymbol{n}, \boldsymbol{nx} \le \boldsymbol{T}$;
$Index = 0$;
For $i = 1 : N_S$
  Use equation (1) to generate $t_{critical}(i)$;
  If $t_{critical}(i) < T$;
    $t = t_{critical}$;
    Based on $x, T, t_{critical}, a_{th}$, calculate the rupture time $t_{rup}(i)$;
    For $k = 1 : n$
      If $t_{rup}(i) > T$
        $Index = Index$;
      else if $a_{th} > a_{in}(k) > y_{th} > a_{in}(k-1)$
        $Index = Index$;
      Else if $a_{in}(k) > a_{th} > a_{in}(k-1)$, and $a_{in}(k-1) < y_{th}$;
        $Index = Index + 1$;
      End
    End
  End
End
End
$p_{rup.tube} = Index / N_s$.

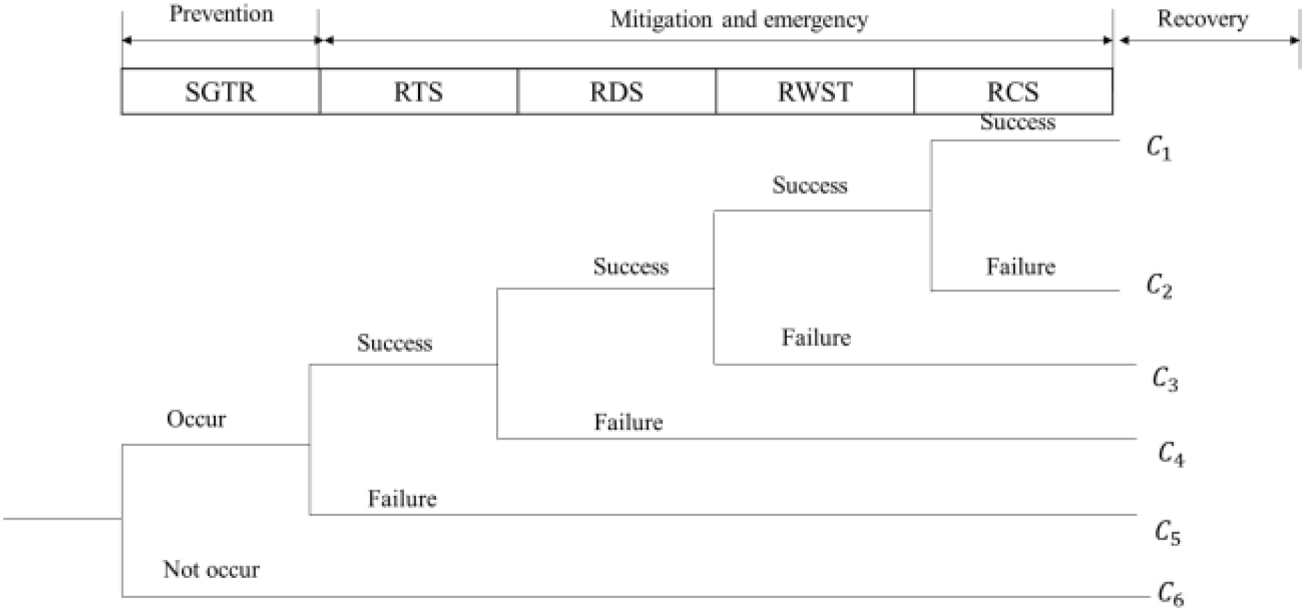


Fig. 2. Schematic ET model of SGTR accident [8].

Table 4.
Classification of consequences.

Consequences	Group	Meaning
C_6	C_I	No SGTR occurs, the NPP is operating normally.
C_1	C_{II}	SGTR occurs, but the consequence is successfully controlled by the mitigation and emergency barriers. The power generation business is temporarily terminated.
$C_2 - C_5$	C_{III}	Core damage is caused by SGTR, the power generation business is terminated for a long time.

Table 5.
Failure probabilities of mitigation measures [9].

Item	Failure probability
RTS ($p_{1,\ b}$)	5.2×10^{-2}
RDS ($p_{2,\ b}$)	1.8×10^{-1}
RWST ($p_{3,\ b}$)	2.4×10^{-1}
RCS ($p_{4,\ b}$)	5.6×10^{-2}

Table 5 summarizes the values of the parameters used in this work, which is set based on the reference [9]. It is worth mentioning that uncertainties in the failure probabilities are not considered here, because these values are just applied to exemplify the applicability of the proposed joint optimization model.

When the SGTR event occurs, there are potential two performances of RTS. If RTS works successfully, evolution of SGTR will depend on the RDT. In this paper, we assume that the safety barriers are binary and independent from each other. Therefore, the failure of a safety barrier is defined as failing to fulfill its function when it is needed. Further, it is assumed that the consequence is determined by the states of the safety barriers based on the event tree model in Fig. 2. The mitigation of core damage was affected by the depressurization by RDS even if the RCS operates normally [33]. Conversely, there are scenarios where core damage is prevented even if some strategies are not normally operated. These phenomena are not considered in this paper, as we assume that the event tree model is complete, i.e., covers all the possible scenarios.

The performance of the mitigation and emergency barriers is primarily determined by their failure probabilities. One common way to reduce the failure probabilities of these barriers is to add redundancies: if one safety barrier collapses, the redundant component/system will substitute it. Redundancy allocation is an optimization problem used to determine the optimal number of redundant components/systems under constraints of costs and resources [34, 35].

In this paper, we consider redundancy allocation optimization using the same type of equipment for the four mitigation and emergency safety barriers. It is easy to show that the failure probability of the i—th measure is:

$$p_i = (p_{i,b})^{n_i+1}, \tag{9}$$

where $p_{i,\ b}$ is the basic failure probability of the i—th safety barrier system (as presented in Table 5) and n_i is the number of redundant systems added to the original one. The total cost associated with the redundancy solution is:

$$C_M = \sum_{i=1}^{4} C_{R,i} \cdot n_i, \tag{10}$$

where $C_{R,\ i}$ is the price for adding one i—th redundancy equipment.

3.3. Modeling recovery measures

Recovery ability refers to the ability of a system to be repaired and quickly restored to its normal operation after failures or disruptions [36]. The repair ability depends on a variety of factors including the training and preparedness of repair groups, the readiness of repair materials and resources, etc. A range of models have been proposed in the literature for modeling the post-disruptive event recovery process [37–39]. According to these models, the performance of the recovery process directly depends on the resources spent on the recovery processes, e.g., the investment on training repair crews, preparing equipment and spare parts used for repairing the failed items [40]. In general, the more budgets or resources planned for recovery, the better the recovery performances.

In this work, we use the following model for the recovery process [41]:

$$T_{recv,i} = \frac{T_{recv,C_i}}{1 + \ln(1 + c_e \cdot \mu \cdot C_R)} \tag{11}$$

where $T_{recv,\ i}$ is a random variable that represents the time needed to recover from the i−th consequence; T_{recv,C_i} denotes the basic recovery time for consequence C_i, which is dependent on the basic requirement on recovery time, C_R denotes the resources invested on the recovery process and c_e is the effective parameter of resources for consequence C_I; μ denotes the relationship between the cost-effectiveness parameters of C_{II} and C_{III} (shown in Table 6); its value should be set by decision makers based on the capability of the organization.

As most literature works (e.g., [5, 42, 43]) did, we also assume that T_{recv,C_i} follows a lognormal distribution:

$$f\left(T_{recv,C_i}\right) = \begin{cases} \frac{1}{\sqrt{2\pi}\beta_i T_{recv,C_i}} e^{\frac{(\ln(T_{recv,C_i}) - \varepsilon_i)^2}{2\beta_i^2}}, & T_{recv,C_i} > 0 \\ 0, & T_{recv,C_i} \le 0 \end{cases} \tag{12}$$

where $f(\cdot)$ is the probability density function (PDF) of T_{recv,C_i};ε_i and β_i are the mean value and standard deviation value of the lognormal distribution, respectively. The values of ε_i and β_i depend on the recovery ability of the target organization solution given the severity of the consequence: it should be noted that the more severe consequence i, the smaller c_e.

It should be noted that for simplification and illustration purposes, we only consider a spontaneous SGTR in this paper. By making this assumption, the recovery process only involves repairing the damages facilities caused by the SGTR. In practice, SGTR can be pressure-induced or temperature-induced leading to sever accident. On these occasions, recovery process not only includes repair, but also environmental cleanup which will lead to additional cost in the BCV calculation process.

4. Joint optimization model based on business continuity

In this section, we develop a joint optimization model to ensure global optimal performance of the safety barriers (prevention, mitigation, emergency and recovery). The joint optimization model is based on the objective of maximizing the business continuity of the plant. In Section 4.1, we recall the business continuity model and the numerical metrics used to guide the optimization. The joint optimization model is presented in Section 4.2. Section 4.3 discusses how to solve this joint optimization problems by using a MIGA. The results of the application on the case study of Section 2 are presented and discussed in Section 4.4.

4.1. Basics of business continuity modeling and assessment

As explained in Section 1, BCM is a comprehensive paradigm that integrates pre-event and post-event management to ensure the resilience and continuous operation of the business. Compared to conventional risk analysis, business continuity assessment not only focuses on the potential hazards and their impacts, but also considers how to mitigate the consequences and quickly recover from the disruption. A quantitative index, i.e., expected business continuity value (EBCV) has been introduced in [5] for business continuity modeling and assessment:

$$EBCV = 1 - \frac{E[L([0,\ T])]}{L_{tol}} \tag{13}$$

where T is the evaluation time horizon for the business continuity assessment; $L([0,\ T])$ is a random variable that describes potential losses in $[0,\ T]$ caused by the disruptive event; L_{tol} denotes the maximum tolerable losses that an organization can tolerate: beyond that level of loss, it will have difficulty to recover the business. As can be seen from this definition, EBCV takes values in $(-\infty,\ 1]$ and a higher EBCV indicates higher business continuity and a EBCV less than zero indicates that on average, the system might have troubles recovering from the disruptions in the considered horizon T.

Two kinds of losses are considered when calculating $L([0,\ T])$: direct loss $L_d([0,\ T])$ and indirect loss $L_{in}([0,\ T])$. The former one represents the losses that are caused directly by the disruptive event, including structural damage of the system. The latter is the revenue loss suffered during the shutdown of the plant. For example, in the case study of SGTR in Section 2, an example of $L_d([0,\ T])$ is the direct financial loss due to damages caused to the assets; $L_{in}([0,\ T])$ might be the downtime costs of the NPP due to the maintenance and recovery process. The total loss is calculated as:

$$L([0,\ T]) = L_{in}([0,\ T]) + L_d([0,\ T]) \tag{14}$$

The direct losses $L_d([0,\ T])$ are mainly determined by the performance of the prevention, mitigation and emergency barriers, whereas the indirect losses $L_{in}([0,\ T])$ are related to the performance of the recovery process. Therefore, the EBCV can be viewed as a global performance measure that integrates the performance of prevention, mitigation, emergency and recovery barriers.

4.2. The joint optimization model

In this section, we present the joint optimization model of the safety

Table 6
Parameter values used in the case study [9].

Parameter	Meaning	Value
L_{tol}	Tolerable losses	5×10^6(€)
L_d	Direct losses	5×10^6(€)
C_{insp}	Cost for one inspection	500 (k€)
$C_{R,\ 1}$	Cost for adding one redundant RTS (Eq. (10))	10 (k€)
$C_{R,\ 2}$	Cost for adding one redundant RDS	4 (k€)
$C_{R,\ 3}$	Cost for adding one redundant RWST	11 (k€)
$C_{R,\ 4}$	Cost for adding one redundant RCS	5 (k€)
C_{plug}	Cost for plugging one tube	5 (k€)
$c_{e,\ II}$	Cost effectiveness parameter for consequence C_{II}	0.001
μ	Relationship value of cost-effectiveness parameter between C_{II} and C_{III}, $c_{e,III} = \mu \cdot c_{e,II}$	0.5
$T_{recv,\ 1}$	Basic recovery time for consequence C_{II}	Lognormal(3.9828, 0.4724)(days)
$T_{recv,\ 2}$	Basic recovery time for consequence C_{III}	Lognormal(6.5922, 0.4724)(days)
T	Business continuity assessment time horizon	10 years
P_e	Unit price for electricity	50 (k€/MWh) [46]
$n_{L,\ 1}$, $n_{L,\ 2}$, $n_{L,\ 3}$, $n_{L,\ 4}$	Lower bound of mitigation measures' number	0
$n_{U,\ 1}$, $n_{U,\ 2}$, $n_{U,\ 3}$, $n_{U,\ 4}$	Upper bound of mitigation measures' number	4
C_{th}	Total budget (Eq. (18))	8000 (k€)

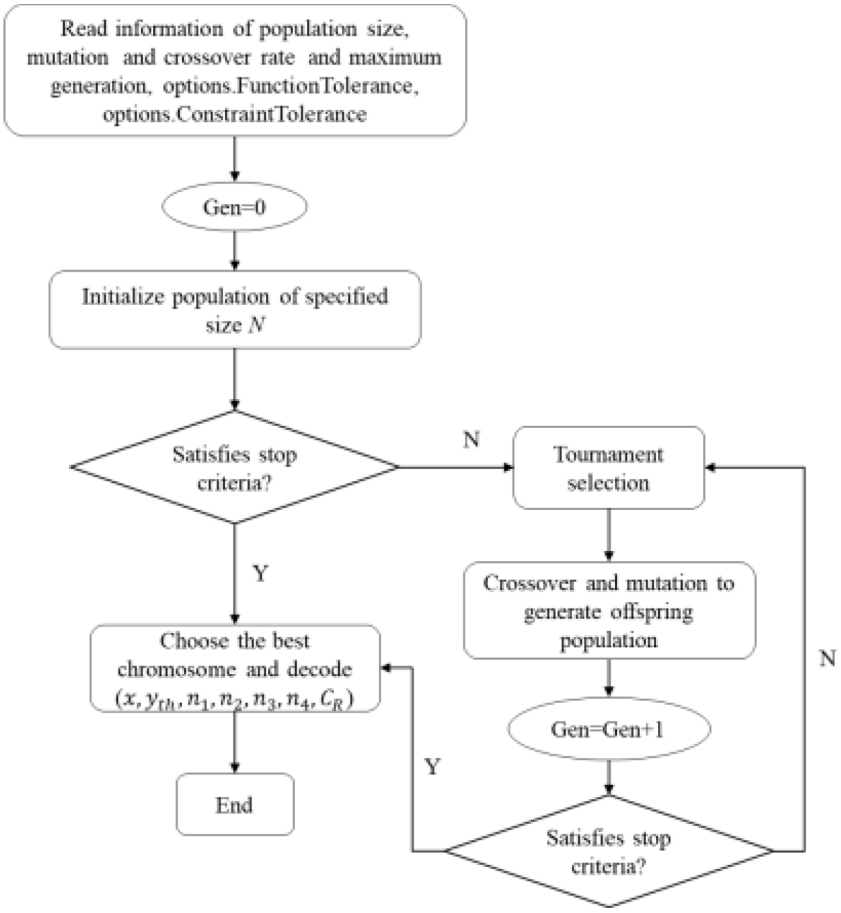


Fig. 3. Flowchart of MIGA.

barriers in different phases, with the objective of maximizing the EBCV [5]. As shown in Eq. (13), the EBCV is determined by the direct loss L_d and indirect losses L_{in} caused by the SGTR-initiated accident. The different possible consequences can be modelled by the ET model in Fig. 2. Based on their severity, these consequences are grouped into the three categories of Table 4. It is assumed that for each category, the NPP suffers a given amount of direct losses, e.g., from equipment damage on the steam generator tube as well as the other parts of the NPP (detailed values are given in Table 6).

If the consequences C_{II} and C_{III} occur, the NPP will become temporality unavailable to produce electricity, until the repair crew successfully handles the incident/accident and restores the normal operation of the NPP. The indirect losses caused in the recovery process can, then, be modelled by:

$$L_{In,C_i} = P_e \cdot C \cdot T_{recv,i}, \tag{15}$$

where L_{in,C_i} represents the indirect losses in the recovery process for consequence C_i, i = I, II,III (L_{in,C_I}=0 as there is no disruption for this event sequence); P_e is the unit electricity price; C is the generating capacity of the NPP; $T_{recv,\ i}$ denotes the recovery time for the i−th consequence and is calculated by Eq. (11). Then, the *EBCV* can be formulated by:

$$\begin{aligned} EBCV &= \frac{\sum_{i=1}^{3} E(L_{C_i}([0,T])) \cdot p_{C_i}}{L_{tol}} \\ &= \frac{\sum_{i=1}^{3} E(L_{d,C_i} + L_{in,C_i}) \cdot p_{C_i}}{L_{tol}} \\ &= [p_{C_{II}} \cdot (L_{d,C_{II}} + P_e \cdot C \cdot E(T_{recv,II})) + p_{C_{III}} \cdot (L_{d,C_{III}} + P_e \cdot C \cdot E(T_{recv,III}))]/L_{tol}. \end{aligned} \tag{16}$$

In Eq. (16), $p_{C_{II}}$ and $p_{C_{III}}$ represent the occurrence probabilities of consequences II and III, respectively. These probabilities are calculated using the ET model in Fig. 2, where the event probabilities in the ET are further determined based on the model developed in Sections 3.1 and 3.2. The distributions of $T_{recv,\ II}$ and $T_{recv,\ III}$ are determined based on Eqs. (11) and (12). $E(T_{recv,\ II})$ denotes the expected recovery time under the II_{th} consequence. The value of $E(T_{recv,\ II})$ is obtained by randomly generated value from the PDF of $T_{recv,\ II}$ in Eq. (12).

As shown in Eq. (16), the EBCV can be conceptually represented as a function of x, y_{th}, n_1, n_2, n_3, n_4, C_R, where x denotes periodic inspection time, y_{th} is preventive maintenance threshold, n_1, n_2, n_3, n_4 are the numbers of redundant components in RTS, RDS, RWST, RCS systems, respectively, C_R represents the resource/budget allocated on the recovery phase. Hence, a joint optimization model can be set up to maximize EBCV:

$$\max \text{EBCV} = f(x, y_{th}, n_1, n_2, n_3, n_4, C_R) \tag{17}$$

$$s.\ t.\ C_P + C_M + C_R \le C_{th}, \tag{18}$$

$$C_P, C_M, C_R \ge 0, \tag{19}$$

$$x \in [6, 7, 8\cdots, 17, 18], \tag{20}$$

$$p_{plug} \le p_{th}, \tag{21}$$

$$\begin{aligned} &n_{L,1} \le n_1 \le n_{U,1},\ n_1 \in N \\ &n_{L,2} \le n_2 \le n_{U,2},\ n_2 \in N \\ &n_{L,3} \le n_3 \le n_{U,3},\ n_3 \in N \\ &n_{L,4} \le n_4 \le n_{U,4},\ n_4 \in N. \end{aligned} \tag{22}$$

The Monte Carlo simulation in Algorithm 1 is used to evaluate the objective function. The first constraint in Eq. (18) regards the total

budget on all safety measures: the total cost on the different safety measures cannot exceed a limited value C_{th}. In Eq. (18),the costs C_P, C_M, C_R are further calculated by Eqs. (6), (10) and (11). The constraint in Eq. (20) defines the possible value of inspection intervals (in months). In this work, it is assumed that the inspection can reveal the exact degradation state of the tube. The constraint in Eq. (21) means that the total number of plugged tubes cannot exceed a maximum value. The value of p_{th} is determined based on the power generation efficiency requirement of the NPP. According to nuclear regulation, a SG of the type considered can tolerate up to 30% plugged tubes before a significant reduction in efficiency occurs [44]. Therefore, we set $p_{th} = 0.3$. The last constraint in Eq. (22) describes the minimal and maximal number of redundant systems for the mitigation measures. The parameter values used in the model are tabulated in Table 6. The parameters applied in Table 6 are referred from the work [9]. In this reference, the parameter values related to cost were set at the same order as their real values. The parameters related to the repair time distribution was set based on [45].

The mitigation and emergency safety barriers considered in this paper is general safety barriers for loss of coolant that are needed anyway. This fact was considered in the optimization model as there were constraints in the model ensuring for each of these safety barriers, a minimum of one barrier is needed (see Eq. (22)). In Eq. (22), n_1, n_2, n_3, n_4 represent the number of redundant safety systems. Even when $n_1 = 0$, we still have one safety barrier designed in the system.

4.3. Mixed integer genetic algorithm

The joint optimization model in Eq. (17) is a mixed integer programming problem, as some decision variables (n_1, n_2, n_3, n_4) are restricted to take integer values whereas the others can take also non-integer values. There are a lot of methods for solving the mixed-integer programming problem, e.g., branch and bound technique [47], Lagrange multiplier [48]. In this paper, we choose the MIGA to solve the joint optimization model for its powerful capability to handle highly complex, nonlinear numerical models and its successful application in related areas like optimization of critical infrastructure resilience [22, 41]. A flowchart of implementing the MIGA is shown in Fig. 3. In this paper, we use the MIGA toolbox in MATLAB 2017b to solve this joint optimization model.

The parameter values of MIGA used in this work are shown in Table 7.

Table 7
Parameter values of MIGA.

Parameter	Values
Population size	40
Crossover rate	0.9
Mutation rate	0.5
Maximum generation	300

4.4. Results and discussions

The optimization model in Eq. (17) was solved numerically using the MIGA described in Section 4.3. The MIGA was run 10 times since the MIGA tends to converge to local minimum. The optimal EBCV value for each run is shown in Fig. 4. As a comparison, the EBCV values from the individual optimization models are obtained by Eq. (17), assuming that all the budget is invested on prevention safety barriers. It can be seen that, in general, the joint optimization works better in terms of achieving higher EBCV than the individual optimization model. More specifically, the joint optimization method can reduce the potential total losses and achieve a higher business continuity against SGTR events than only investing all the budget on the prevention phase. The same conclusions are revealed considering the cost of investing only in the mitigation and emergency phase or in the recovery phase.

To further examine the differences between the different optimization strategies, we compare the optimal values of the decision variables from different optimization models in Table 8. Note that for the joint optimization model, the best solution among the 10 times is selected as the optimal solution to the model: it requires a periodical inspection of the steam generation tube every 15 months and a tube should be plugged whenever its crack length exceed 7.8780 mm. The number of redundant components in mitigation and emergency barriers are $n_1 = 1$, $n_2 = 4$, $n_3 = 4$, $n_4 = 4$, where $1 \sim 4$ correspond to the redundant components in the RTS, RDS, RWST, RCS, respectively. Another $C_R = 108.2384$(k€) is allocated for improving the performance of the recovery process.

In Fig. 5. we compare the performances of the different safety barriers under different optimization model. In particular, the probability of SGTR is used to represent the performance of prevention safety barriers: a higher value of SGTR probability indicates worse performance of the prevention barriers. The failure probability of mitigation indicates the mitigation system performance, where the higher the failure probability the worse the performance of the mitigation system. Moreover, the mean time to restore normal operation and the PDF of the recovery times show the recovery ability, where longer recovery times indicate poorer recovery ability.

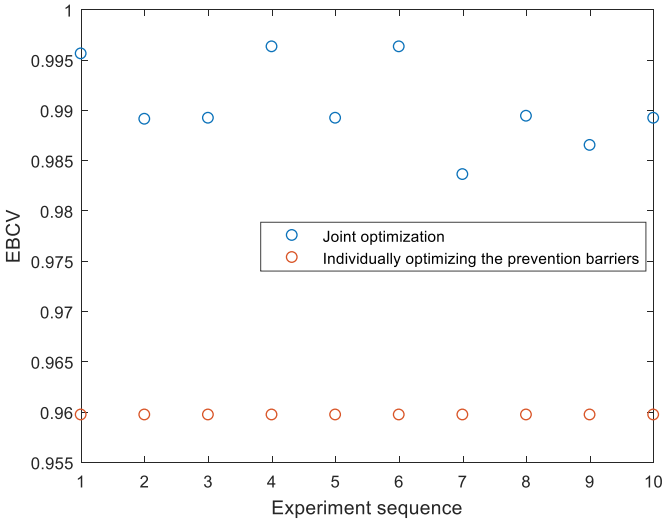


Fig. 4. Comparison between the proposed joint optimization and individually optimizing the prevention barriers (C_{total} = 8000k€).

It can be seen from Fig. 5(a) that the SGTR occurrence probability takes the lowest value $6.9998 \times 10^{-4}$when all the budget is spent on the preventive measures. However, the corresponding EBCV is lower than the joint optimization model (as shown in Table 8), indicating that the joint optimization model can achieve a better performance globally. This is because considering the preventive measures individually might sometimes affect the performance of the mitigation and emergency and recovery processes, as less resources can be invested on these measures. Similar results can be found in Fig. 5(b), (c) and (d): although the solution obtained from the joint optimization model is not optimal with respect to each safety barrier, it can achieve an overall optimal performance with respect to business continuity. It is worth mentioning that this is the only acceptable solution as the other severe accidents would be unacceptable in terms of "losses".

5. Sensitivity analysis

A sensitivity analysis is conducted to investigate how does the

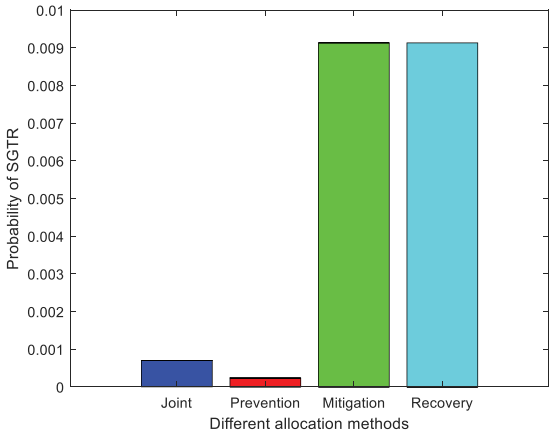
Table 8
Comparison results for business continuity under different strategies.

Variable	Joint optimization	Prevention measures only	Mitigation and emergency	Recovery (only)
x	1.25 (year)	1 (year)	/	/
y_{th}	7.8780 (mm)	14.1347(mm)	/	/
n_1	1	/	4	0
n_2	4	/	4	0
n_3	4	/	4	0
n_4	4	/	4	0
C_P	7792.2 (k€)	7990.05(k€)	0	0
C_M	90 (k€)	0	120 (k€)	0
C_R	108.2384 (k€)	0	0	8000 (k€)
C_{th}	7990.4 (k€)	7999.05 (k€)	120 (k€)	8000 (k€)
EBCV	0.9963	0.9597	−20.06	−44.63

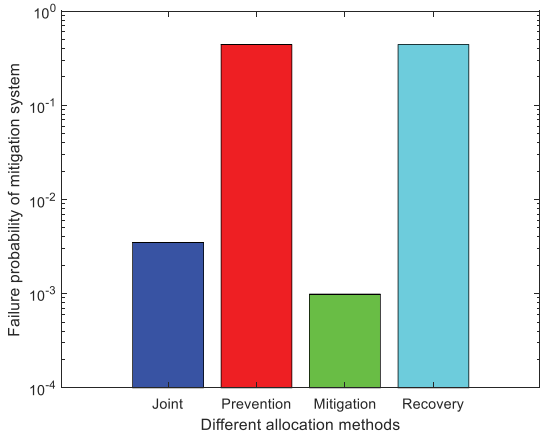
optimal design solution changes with total budget, basic failure probability of mitigation barriers ($p_{i,\ b}$ in Eq. (9)) and cost-effectiveness parameter (c_e in Eq. (11)) of improving recovery barriers. The sensitivity analysis is gradually done by changing the parameter of interest while fixing the values of the other parameters.

Fig. 6 shows how does the optimal EBCV change under different values of total budget C_{th}. As expected, as the total budget grows, the optimal EBCV correspondingly increases. The error bar in Fig. 6 shows the mean and standard deviation of EBCV using the proposed joint optimization model for 10 experiments. The main reason for this is that as the C_{th} increases, more available resources can be used to enhance system business continuity. However, it should be noted that when C_{th} exceeds 8000k€, the marginal effects on EBCV become very small. This result shows that although increasing C_{th} can improve the business continuity, increasing the budget further when it already reaches a threshold value (8000k€ in this case) might become a waste of resources, as it cannot further improve the business continuity. In practice, the most cost-benefit way of setting C_{th} is to set the budget around this threshold value. A dramatic change of the EBCV value can also be observed in Fig. 6. This is mainly because when the total budget is larger than 7888.2k€, the occurrence probability of SGTR will dramatically decreased.

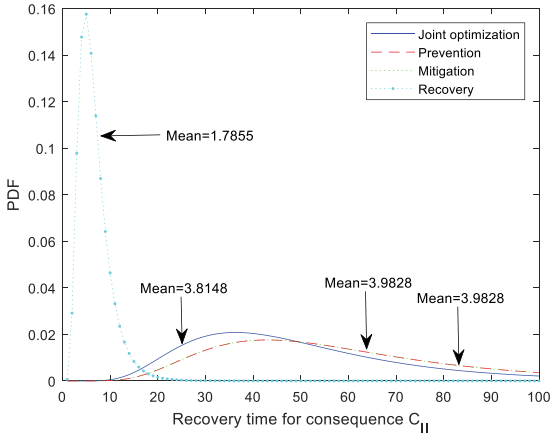
To investigate the influence of the cost-effective parameter c_e on the optimal solution, the optimal values of EBCV under different values of c_e are shown in Fig. 7. Fig. 7 shows the joint optimization results where the C_{th} is 7500k€ and 8000k€, respectively. As can be seen from Fig. 7, the optimal EBCV value is very sensitive to the increase of c_e when the total budget is small (7500k€), whereas when the C_{th} is large enough, the optimal EBCV value almost does not change with the changing of c_e. This is because the partial differentiation of $T_{recv.i}$ with respect to C_R is:



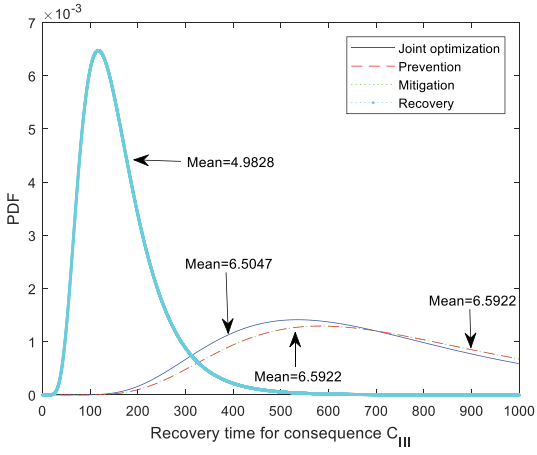
(a) Probability of SGTR.



(b) Failure probability of mitigation system.



(c) PDF of recovery time for consequence C_{II}.



(d) PDF of recovery time for consequence C_{III}.

Fig. 5. Performance indexes in prevention, mitigation & emergency, and recovery phases.

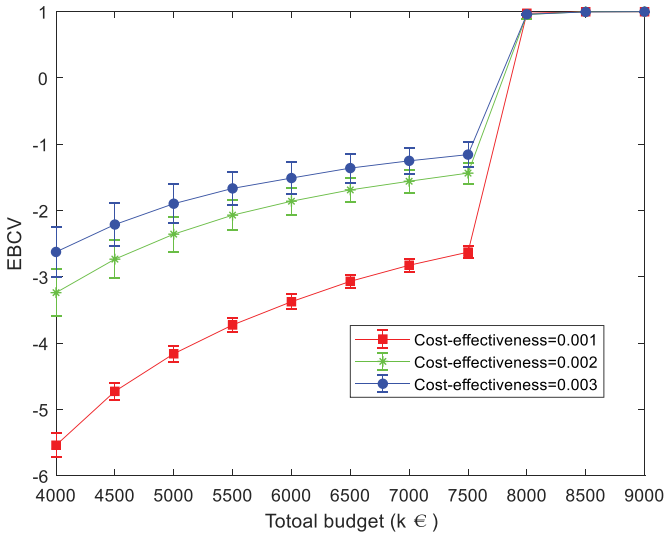


Fig. 6. Sensitivity of the optimal EBCV against the C_{th}.

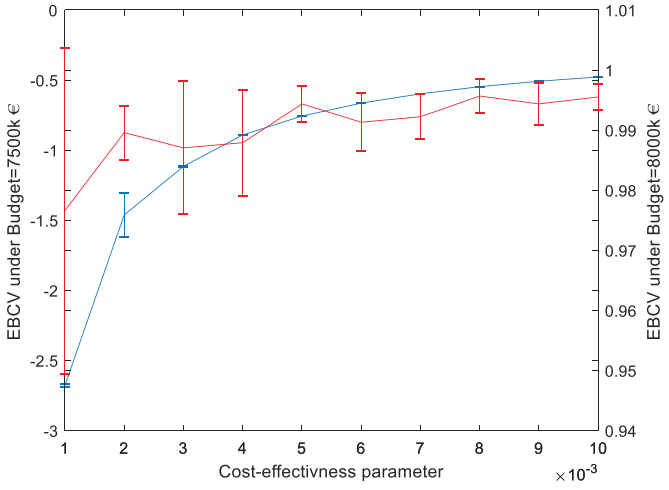


Fig. 7. Comparision EBCV with different cost-budgets.

$$\frac{\partial T_{recv,i}}{\partial C_R} = \frac{-c_e}{c_e \cdot C_R + 1} \tag{23}$$

We can see that a smaller value of C_R leads to a larger value of $\frac{\partial T_{recv,i}}{\partial C_R}$: therefore, the smaller the total budget (corresponding to the smaller C_R), the more sensitive is EBCV to the value of c_e.

To investigate the influence of the basic failure probabilities of mitigation barriers on the budget allocated to improve them, we do a sensitivity analysis by changing the mitigation failure probability from 70% to 130% of the baseline values used in Table 2 for the failure probability of the mitigation barriers. As shown in Fig. 8 (where the black lines are the minimum and maximum values among the different runs of the MIGA, the red line denote the median values, the blue box shows the upper 75% quartile and lower 25% quartile, respectively, when the failure probabilities of the mitigation barriers increase, the proportion of the total budget that is allocated for improving the mitigation barriers increases accordingly, This is a straightforward conclusion: more redundant safety systems are needed if the mitigation safety system has higher failure probability.

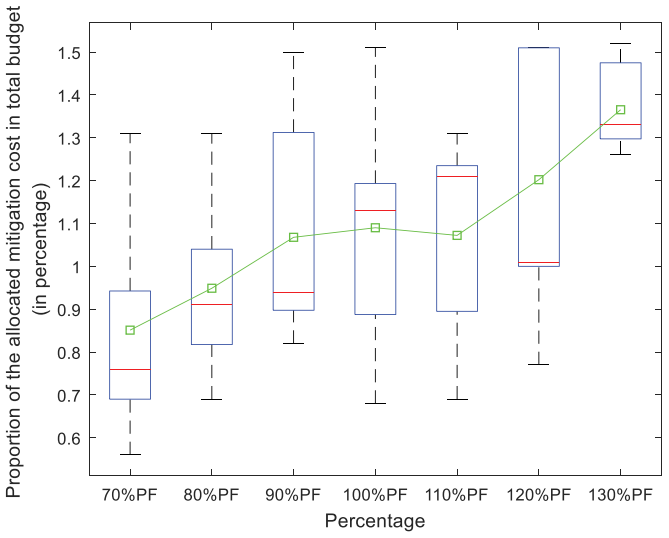


Fig. 8. Schematic of changing failure probability of mitigation measures (70%−130%) under budget C_{total} = 8000k€..

To investigate the influence of direct loss cost on the EBCV, we conduct a sensitivity analysis by changing the direct loss cost from 1% to 10% of the baseline value used in Table 6, where c_e = 0.001.As shown in Fig. 9 (where the black lines are the minimum and maximum values among the different runs of the MIGA, the red line denote the median values, the blue box shows the upper 75% quartile and lower 25% quartile respectively), when the direct loss cost increases, the EBCV decreases accordingly. This is a straightforward conclusion: the more direct loss cost, the less continuity of the system would have. Moreover, the outlier values in Fig. 9 are caused by the local optimum of GA.

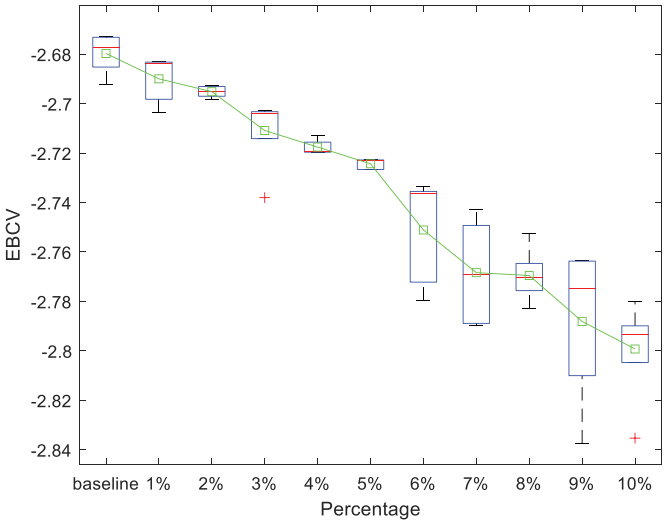


Fig. 9. Schematic of changing direct loss cost (1%−10%) under budget C_{total} = 7500k€..

In Fig. 10, we also investigated how the proportion of the allocated mitigation costs changed with the increased direct loss costs. As shown in Fig. 10, the proportion of the allocated mitigation costs does not change significant with the increase of direct loss costs (compared to the uncertainty in the optimization). This is because, even when the

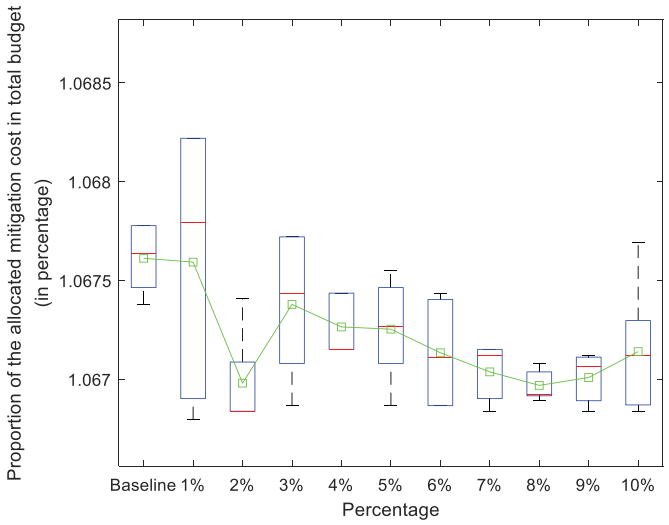


Fig. 10. Schematic of changing direct loss cost (1% − 10%) under budget $C_{total} = 7500k€$..

direct loss cost takes its baseline value, the proportion of the mitigation cost has been very close to its theoretical maximal value, which is obtained when all the safety barriers implement maximal degrees of redundancy.

6. Discussion and conclusions

In this work, the proposed joint optimization method is applied on a NPP. It can be naturally extended to application in other industrial systems. For example, in oil and gas industry, there are different safety measures against lighting attacks on oil tanks: prevention measures include lighting protection masts; mitigation and emergency measures include automatic rim seal fire extinguishing system, fixed foam system and fire brigade; while recovery measures are restore and/or replace the damaged tanks. The proposed joint optimization model can provide a holistic framework for optimizing the safety measures under limited budgets or resources.

A major limitation of the developed method is that to implement the optimization, it is required that the failure models and its parameters are known a priori. In practice, we often do not have precise values for model parameters. A potential solution is to consider dynamic parameter updating approaches based online-collected condition monitoring data.

Another limitation of the developed method is the event tree modeling for mitigation and emergency phases. This simplified analysis might not fully capture the complex processes occurring in a realistic mitigation and emergency response following a disruption. For example, in the event tree model considered in this work, the use of auxiliary feed water to cool the core is not modeled. Further, in practice, complex interactions might arise among the safety barriers [9], which questions the validity of using event trees that assume independence among safety barriers. In the future, more realistic models can be investigated for the disruption processes.

In this work, a joint optimization model is formulated to optimally allocate the limited resources among different safety barriers to enhance business continuity of NPPs exposed to the risk of SGTR-initiated accordingly. The model allows integrated design of prevention, mitigation, emergency and recovery barriers. The prevention barriers are modeled as periodical inspection and maintenance based on a physics-of-failure model. The mitigation and emergency barriers are modelled through a parallel redundancy model. The recovery measures are described by a logarithmic recovery function. An application demonstrates the feasibility of the developed approach that can provide a globally optimal solution superior to that obtained by optimizing the safety barriers separately. Through sensitivity analysis, we found out that: (1) larger resource budget can result in higher business continuity; the change of EBCV is marginally decreasing with the increase of the budget; there is an optimal budget for the given NPP; (2) the higher failure probability of the safety measures in mitigation phase, the higher the proportion of the total budget that is allocated for improving the mitigation measures; (3) the smaller the budget, the more sensitive to cost-effectiveness parameter is business continuity.

CRediT authorship contribution statement

Jinduo Xing: Writing - original draft, Investigation, Visualization. **Zhiguo Zeng:** Methodology, Validation, Writing - review & editing. **Enrico Zio:** Conceptualization, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The work of Dr. Jinduo Xing is supported by China Scholarship Council (No. 201506450020).

References

[1] Riznic J. Steam generators for nuclear power plants. Elsevier Science; 2017.
[2] Kim H, Kim JT, Heo G. Failure rate updates using condition-based prognostics in probabilistic safety assessments. Reliab Eng Syst Saf 2018;175:225–33.
[3] Chatterjee K, Modarres M. A probabilistic physics-of-failure approach to prediction of steam generator tube rupture frequency. Nuclear Sci Eng 2012;170(2):136–50.
[4] Sklet S. Safety barriers: definition, classification, and performance. J Loss Prev Process Ind 2006;19(5):494–506.
[5] Zeng Z, Zio E. An integrated modeling framework for quantitative business continuity assessment. Process Saf Environ Protect 2017;106:76–88.
[6] Ren F, Zhao T, Jiao J, Hu Y. Resilience optimization for complex engineered systems based on the multi-dimensional resilience concept. IEEE Access 2017;5:19352–62.
[7] ISO, ISO 17776 Petroleum and natural gas industries – Offshore production installations – Major accident hazard management during the design of new installations, 2016. p. 96.
[8] Di Maio F, Antonello F, Zio E. Condition-based probabilistic safety assessment of a spontaneous steam generator tube rupture accident scenario. Nuclear Eng Des 2018;326:41–54.
[9] Zeng Z, Zio E. Joint optimization of business continuity by designing safety barriers for accident prevention, mitigation and emergency responses. Proceedings of the 3rd international conference on system reliability and safety (ICSRS). IEEE; 2019.
[10] Losada C, Scaparra MP, O'Hanley JR. Optimizing system resilience: a facility protection model with recovery time. Eur J Oper Res 2012;217(3):519–30.
[11] Wade KC. Steam generator degradation and its impact on continued operation of pressurized water reactors in the United States. Energy Inf Adm Electric Power Monthly 1995;66.
[12] Sato A, Lyamzina Y. Diversity of concerns in recovery after a nuclear accident: a perspective from Fukushima. Int J Environ Res Public Health 2018;15(2):350.
[13] Mancuso A, Compare M, Salo A, Zio E. Portfolio optimization of safety measures for the prevention of time-dependent accident scenarios. Reliab Eng Syst Saf 2019:106500.
[14] Jovanovic A. Risk-based inspection and maintenance in power and process plants in Europe. Nuclear Eng Des 2003;226(2):165–82.
[15] Yang R, Kang J, Quan Z. An enhanced preventive maintenance optimization model based on a three-stage failure process. Sci Technol Nuclear Install 2015;2015.
[16] Fouladirad M, Grall A, Dieulle L. On the use of on-line detection for maintenance of gradually deteriorating systems. Reliab Eng Syst Saf 2008;93(12):1814–20.
[17] Yang J-E, Hwang M-J, Sung T-Y, Jin Y. Application of genetic algorithm for reliability allocation in nuclear power plants. Reliab Eng Syst Saf 1999;65(3):229–38.
[18] Peiravi A, Karbasian M, Ardakan MA, Coit DW. Reliability optimization of series-parallel systems with K-mixed redundancy strategy. Reliab Eng Syst Saf 2019;183:17–28.
[19] Coit DW, Smith AE. Reliability optimization of series-parallel systems using a genetic algorithm. IEEE Trans Reliab 1996;45(2):254–60.
[20] Pham H, Galyean WJ. Reliability analysis of nuclear fail-safe redundancy. Reliab Eng Syst Saf 1992;37(2):109–12.
[21] Yumashev D, Johnson P, Thomas PJ. Economically optimal strategies for medium-term recovery after a major nuclear reactor accident. Process Saf Environ Protect

2017;112:63–76.
[22] Ouyang M, Wang Z. Resilience assessment of interdependent infrastructure systems: with a focus on joint restoration modeling and analysis. Reliab Eng Syst Saf 2015;141:74–82.
[23] Figueroa-Candia M, Felder FA, Coit DW. Resiliency-based optimization of restoration policies for electric power distribution systems. Electr Power Syst Res 2018;161:188–98.
[24] ISO. ISO 22301, in societal Security- Business Continuity management Systems-Requirements. Switzerland: International Organization for Standardization; 2012.
[25] Hassel H, Cedergren A. Exploring the conceptual foundation of continuity management in the context of societal safety. Risk Anal 2019;39(7):1503–19.
[26] Xing J, Zeng Z, Zio E. Dynamic business continuity assessment using condition monitoring data. Int J Disas Risk Reduct 2019:101334.
[27] Rebello S, Yu H, Ma L. An integrated approach for real-time hazard mitigation in complex industrial processes. Reliab Eng Syst Saf 2019.
[28] Hoseyni SM, Di Maio F, Zio E. Condition-based probabilistic safety assessment for maintenance decision making regarding a nuclear power plant steam generator undergoing multiple degradation mechanisms. Reliab Eng Syst Saf 2019;191:106583.
[29] Blain C, Barros A, Grall A, Lefebvre Y. Modelling of stress corrosion cracking with stochastic processes–application to steam generators. Proceedings of the European safety and reliability conference risk, reliability and societal safety. 2007.
[30] Lewandowski R, Denning R, Aldemir T, Zhang J. Implementation of condition-dependent probabilistic risk assessment using surveillance data on passive components. Ann Nucl Energy 2016;87:696–706.
[31] Buck JA, Underhill PR, Morelli JE, Krause TW. Simultaneous multiparameter measurement in pulsed eddy current steam generator data using artificial neural networks. IEEE Trans Instrum Meas 2016;65(3):672–9.
[32] Narayanan M, Kumar A, Thirunavukkarasu S, Mukhopadhyay C. Development of ultrasonic guided wave inspection methodology for steam generator tubes of prototype fast breeder reactor. Ultrasonics 2019;93:112–21.
[33] Jeon J, Choi W, Kim NK, Kim SJ. Numerical investigation of in-vessel core coolability of PWR through an effective safety injection flow model using MELCOR simulation. Ann Nucl Energy 2018;121:350–60.
[34] Cai B, Xie M, Liu Y, Liu Y, Feng Q. Availability-based engineering resilience metric and its corresponding evaluation methodology. Reliab Eng Syst Saf 2018;172:216–24.
[35] Tohidi H, Chavoshi S, Bahmaninezhad A. A continuous-time Markov chain model for redundancy allocation problem: an economic analysis. Qual Reliab Eng Int 2019.
[36] Cincotta S, Khakzad N, Cozzani V, Reniers G. Resilience-based optimal firefighting to prevent domino effects in process plants. J Loss Prev Process Ind 2019;58:82–9.
[37] Fang Y-P, Zio E. An adaptive robust framework for the optimization of the resilience of interdependent infrastructures under natural hazards. Eur J Oper Res 2019;276(3):1119–36.
[38] Duffey RB, Ha T. The probability and timing of power system restoration. IEEE Trans power Syst 2013;28(1):3–9.
[39] MacKenzie CA, Zobel CW. Allocating resources to enhance resilience, with application to superstorm sandy and an electric utility. Risk Anal 2016;36(4):847–62.
[40] Ouyang M, Dueñas-Osorio L, Min X. A three-stage resilience analysis framework for urban infrastructure systems. Struct Saf 2012;36-37:23–31.
[41] Zhang C, Kong J-j, Simonovic SP. Restoration resource allocation model for enhancing resilience of interdependent infrastructure systems. Saf Sci 2018;102:169–77.
[42] Ferrario E, Zio E. Assessing nuclear power plant safety and recovery from earthquakes using a system-of-systems approach. Reliab Eng Syst Saf 2014;125:103–16.
[43] Arif A, Ma S, Wang Z, Wang J, Ryan SM, Chen C. Optimizing service restoration in distribution systems with uncertain repair time and demand. IEEE Trans Power Syst 2018;33(6):6828–38.
[44] Lewandowski R. Incorporation of corrosion mechanisms into a state-dependent probabilistic risk assessment. The Ohio State University; 2013.
[45] Bonavigo L. M.D.S., issues for nuclear power plants steam generators. Steam generator systems: operational reliability and efficiency. 2011.
[46] Borovkova S, Schmeck MD. Electricity price modeling with stochastic time change. Energy Econ 2017;63:51–65.
[47] Deep K, Singh KP, Kansal ML, Mohan C. A real coded genetic algorithm for solving integer and mixed integer optimization problems. Appl Math Comput 2009;212(2):505–18.
[48] Bertsekas DP. Constrained optimization and lagrange multiplier methods. Academic press; 2014.